

1. Introduction - Why Virtualization?

When we think of a computer, we usually imagine one system being used by one person at a time. Traditionally, if you had a machine, it could run only one operating system and support one main activity or a few related applications. Imagine a server in a company. That single server might be dedicated only for payroll applications, or only for email, or just as a web server. What happens then? We end up buying multiple machines for multiple tasks, and most of those machines use only a portion of their capacity – the rest of the resources sit idle.

Virtualization solves this inefficiency. It allows us to take a single powerful physical computer and divide it logically into multiple smaller computers, called virtual machines. Each virtual machine behaves as though it is a completely separate, independent computer, even though they are all running on the same hardware underneath. With virtualization, it is as if one physical computer becomes many smaller computers. This is the magic that allows companies like Google, Microsoft, and Amazon to rent out slices of their resources to millions of users at the same time.

A student-friendly analogy is to compare it to Netflix. You buy one subscription, and with family sharing, five people can watch five different shows at the same time, on different devices. It looks as though you each have your own Netflix, but ultimately it is coming from a single account. Likewise, virtualization lets multiple users feel as though they each own an independent computing resource.

2. Abstraction and Layered Structure

Another key idea in cloud computing is abstraction. Abstraction means we don't expose all the complicated technical details, only what the user needs to see. For example, when you drive a car, you have a steering wheel, accelerator, and brake. You never bother to know how the fuel injector works, or how the engine fires cylinders. Those details are hidden from you, letting you focus on driving.

Cloud computing uses abstraction in the same way. You log into Gmail, and you don't care in which data center your email is stored or what kind of hard drive it's placed on. All that complexity is hidden behind simple interfaces.

Virtualization makes heavy use of a layered structure. At the bottom, we have the physical hardware – CPUs, memory, disk drives. On top of this sits the hypervisor, which is software that creates virtual machines. Each of those virtual machines in turn can run its own complete operating system independently. At the very top, the end user just sees applications like Google Docs or Zoom.

I like to compare this to a layered cake. The bottom spongy bread is your hardware, the cream in the middle is the hypervisor, the flavored layers of cream above are the virtual machines, and the cherry and decoration on the top are the applications that users interact with. Most people only see the tasty top layer, but every layer beneath is crucial to hold it together.

3. Hypervisors

The hypervisor is the heart of virtualization. Imagine you have one playground and many children want to use it. Someone has to act like a supervisor to ensure every child gets a fair turn at using the swing and slide. Without that supervisor, one strong child may occupy everything, and the others may be left out. Similarly, in a computer, the hypervisor divides the physical resources – CPU cycles, RAM, and disk space – among different virtual machines, making sure they run smoothly without interfering with each other.

There are two types of hypervisor. The first type is called a Type I or bare-metal hypervisor. This hypervisor runs directly on the hardware itself. It does not need a host operating system in between. This makes it very efficient and very fast, so it is mostly used in professional data centers. Examples are VMware ESXi or Microsoft Hyper-V. You can think of this like a direct flight – efficient and quick.

The second type is called a Type II or hosted hypervisor. This one runs on top of an existing operating system, like Windows or Linux, and then creates virtual machines inside it. It is easier to install and great for personal computers and testing purposes. Popular examples are Oracle VirtualBox and VMware Workstation. This is more like a connecting flight with a stopover. It works fine, but it is slightly slower because there is the extra layer of a host operating system in the middle.

4. Virtualization versus Traditional Approach

In order to understand the power of virtualization, let us compare it with the traditional approach. In traditional computing, one machine typically runs only one operating system and one major application. If you wanted more applications, you ended up buying more machines. The problem was that most of these machines were underutilized – their processors and memory were mostly idle, but we still had to keep them running, wasting electricity, money, and space.

Virtualization changes this completely. Now, the same physical hardware can run multiple virtual machines, each with its own different operating system and separate applications. One machine can be Linux-based handling databases, another virtual machine can be Windows-based handling Microsoft applications, and another could be dedicated to testing new software, all running simultaneously on the same hardware.

A simple example is to think of one bus. In the traditional approach, that bus would carry only one passenger. Completely wasteful. Virtualization allows forty passengers to sit in one bus and all travel at the same time. Same infrastructure, but now fully utilized.

5. Types of Virtualization

There are several flavors of virtualization, each targeting a different kind of resource.

The first and most popular is server virtualization. Here, a single powerful server is divided into multiple smaller virtual servers. This is useful when a company needs several servers but doesn't want to physically buy many machines.

Next is network virtualization. This creates multiple "virtual networks" on the same physical cabling. For example, in your college computer lab, there might only be one set of network wires, but you can create separate virtual networks for different classes or projects, all isolated logically.

Then comes storage virtualization. It allows us to combine multiple storage devices into one big logical storage pool. You might think of it like combining many USB flash drives into one giant virtual pen drive that users can access.

Finally, there is desktop virtualization. This lets your desktop run not on your local computer but on a remote server. During online labs, many students may have experienced this when you login and see a Windows desktop running in the browser, even though it is not on your laptop but on a server.

6. Virtualization Environments

Now let us discuss different types of environments that can be built using virtualization. One kind is scheduling-based. Here, the focus is on sharing CPU time fairly. It is like a teacher giving each student two minutes to speak, so everyone gets a fair turn, no one monopolizes.

Another type is load-distribution-based. Here, the system distributes heavy work evenly across many machines. It is just like a restaurant manager making sure customers are spread evenly among waiters.

There is also energy-aware virtualization. Here, the aim is to save power. If fewer users are active, then some machines can be safely switched off. Think of a classroom where only ten students come. The teacher switches off half the lights and fans to save electricity.

Finally, we have transactional-based environments. These ensure very quick and reliable responses to each request. For example, when you withdraw money from an ATM, the transaction must complete almost instantly - you cannot be kept waiting or lose reliability. Transactional-based virtualization environments ensure that happens.

7. Load Balancing in Virtualization

An important companion to virtualization is load balancing. Imagine a pizza shop where ten chefs are available. If all orders go to just one chef, he collapses under pressure, while the other nine remain idle. A smart manager distributes orders across all chefs evenly, so each gets a fair share and all customers are served quickly.

Load balancing in cloud computing works the same way. If too many requests come in, they are distributed among multiple servers or virtual machines. Google Cloud is an excellent real-world example. When millions of people open YouTube or Gmail at the same time, the requests are spread across Google's global network of servers - in India, Singapore, the USA - so that no single server crashes. To us as users, everything looks smooth and simple, but in the background, load balancing ensures fairness and availability.

8. Machine Imaging and OVF

Another powerful concept is machine imaging. This is like taking a snapshot of your entire computer at a given time. Imagine you carefully installed Linux and configured Python, TensorFlow, and some databases. Instead of reinstalling everything again if you need another system, you just save this whole machine as an image. Later, you can recreate that exact machine in minutes. It's like taking a photo of your phone screen and settings. If your phone crashes, you can restore everything as it was.

To make these machine images portable across different platforms, the industry uses a standard format called Open Virtualization Format (OVF). With OVF, a virtual machine created in VMware can be exported and later imported into Oracle VirtualBox. It acts like saving your document as a PDF, so anyone can open it, regardless of whether they use Microsoft Word or LibreOffice.